AI Risk Warnings, Media Tone and AI Equity Returns

Evidence from September 8 to September 30, 2026

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Abstract. On the evening of September 8, 2026, Anthropic alignment lead Evan Hubinger publicly agreed with a resigning colleague that AI could kill all humans and put the probability above 10% within a decade. Using 488 news headlines, 42,486 Hacker News comments, 4,331 Reddit posts, a systematic hand-collected sample of X posts and GDELT web-news aggregates, I find a sharp and short-lived rise in *attention* to existential AI risk rather than a broad shift in AI sentiment: the share of Hacker News comments using extinction-type vocabulary more than doubled on September 9 and fell back within about a week, while overall polarity barely moved. An equal-weight basket of six AI equities returned 2.72% over the window (SPY -0.74%), driven by Meta. Changes in news or social sentiment do not Granger-cause basket returns ($p = 0.26$ and 0.27); neither do changes in risk attention. Coverage is measurably asymmetric: the Guardian is far more negative and risk-focused than TechCrunch, and US web outlets cover AI risk more negatively than non-US outlets, echoing the Covid-19 news study of Sacerdote, Sehgal and Cook (2020).

1. Event and research questions

Late on September 8, Anthropic researcher Jacob Coxon announced his resignation with a warning that AI developers were "gambling with our lives". At 21:27 New York time (the timestamp is encoded in the post identifier) Evan Hubinger replied on X: "Jacob is correct here—we really do earnestly believe AI could kill all humans! I personally think it is >10% within the next decade." Because the post appeared after the close, September 9 is the first session in which markets could respond. Coverage then continued for three weeks: Anthropic's chief executive called for an industry slowdown on September 12, AI agents were reported to have hacked other companies, lab leaders addressed the UN Security Council, and Anthropic's IPO prospectus of September 29 itself warned that its technology could end humanity.

Following the course readings, I ask three questions. (i) Did sentiment about AI change over September 8-30, and how? (ii) Do changes in sentiment Granger-cause returns on a basket of AI equities, as Smailović, Grčar and Žnidaršič report for tweets and Apple returns and Souza et al. (2015) test separately for news and Twitter? (iii) Is there a measurable bias in how media and social media cover AI risk, in the sense of the Sacerdote et al. (2020) study of Covid-19 coverage?

2. Data

Table 1. Data sources, September 8-30, 2026 (New York time).

Source	Content collected	N
Guardian	All headlines in the Artificial Intelligence section archive	226
TechCrunch	All headlines in the AI category (publisher API)	262
Hacker News	All comments containing AI vocabulary (Algolia API, 9 queries)	42,486
Reddit	All posts with AI vocabulary in r/artificial, r/singularity, r/technology, r/stocks, r/investing, r/wallstreetbets (Arctic Shift archive)	4,331
X	Hand-collected: first 20 quotes and 20 replies to the Hubinger post, and the 3 latest posts per day from a keyword search	107
GDELT	English-language web news mentioning "artificial intelligence"; tone histograms and daily volumes, US vs non-US sources	2,199 risk articles
Prices	Adjusted closes, NVDA MSFT AMZN GOOGL META AVGO SPY QQQ (Yahoo Finance)	17 sessions

AI vocabulary is AI, AGI, LLM, ChatGPT, OpenAI, Anthropic, Claude, artificial intelligence, large language model and superintelligence; the search is deliberately broad so that the sample is not pre-selected on fear. Hacker News is a technology forum, the Reddit sample adds two investor communities, and the two news outlets differ in audience: the Guardian is a general-interest UK-based newspaper, TechCrunch a US technology trade publication. Because X's historical search API is not available at reasonable cost, X is represented by a small systematic sample collected by hand: posts were recorded in the order displayed, skipping only advertisements, non-English and media-only posts, and each timestamp was recovered from

the post identifier. GDELT returned no data for seven days in mid- and late September (a collection outage), so it is used mainly for the cross-sectional comparison. As Chen, Lerman and Ferrara (2020) stress for Covid-19 corpora, keyword choice and API coverage shape what such samples contain; deleted posts cannot be recovered. Retrieval URLs, timestamps and hashes for every source are kept with the code.

3. Measuring sentiment

The main score is P(positive) − P(negative) from the Cardiff Twitter-RoBERTa sentiment model (Loureiro et al., 2022), used frozen without any fitting to returns; a text is negative when the negative class is most likely. VADER (Hutto and Gilbert, 2014) and the Hu-Liu (2004) opinion lexicon serve as sensitivity measures, since Carvalho and Plastino (2020) show that results for short social texts depend heavily on the feature set used. Reddit posts were scored one at a time with the identical model file (verified by checksum); on 36 previously scored texts this reproduces the class assignment in 97% of cases. Two transparent indicators track topic rather than tone: *risk vocabulary* (risk, danger, threat, safety, harm, kill, alignment, catastrophe, doom and variants) and *existential vocabulary* (extinction, existential, superintelligence, "kill all/us/every", doom). To check the measures, a stratified random sample of 150 texts (40 Guardian, 40 TechCrunch and 70 Hacker News, half of the latter drawn from comments with risk vocabulary) was coded text by text under a written codebook (sentiment and AI-risk content) without access to model output. RoBERTa matches the three-class coding in 65% of texts (κ = 0.39) against 45% for VADER (κ = 0.18). RoBERTa's negative calls are reliable (precision 78%) but conservative (recall 50%), so negative shares below are lower bounds; the risk-vocabulary flag has precision 84% and recall 59% for genuine AI-risk discussion. I recoded a stratified random 50-text subsample myself, blind to model scores and to the first coding. Agreement with the first coding is 78% for sentiment (Cohen's κ = 0.63) and 82% for AI-risk content (κ = 0.62), substantial agreement on the Landis-Koch scale; most disagreements are texts I judged negative that the first coding left neutral. Against my labels RoBERTa agrees in 52% of cases and VADER in 46%, and RoBERTa's negative calls keep high precision (75%) with lower recall (36%), which strengthens the reading of negative shares as lower bounds.

4. The trend: a spike in risk attention, not a shift in mood

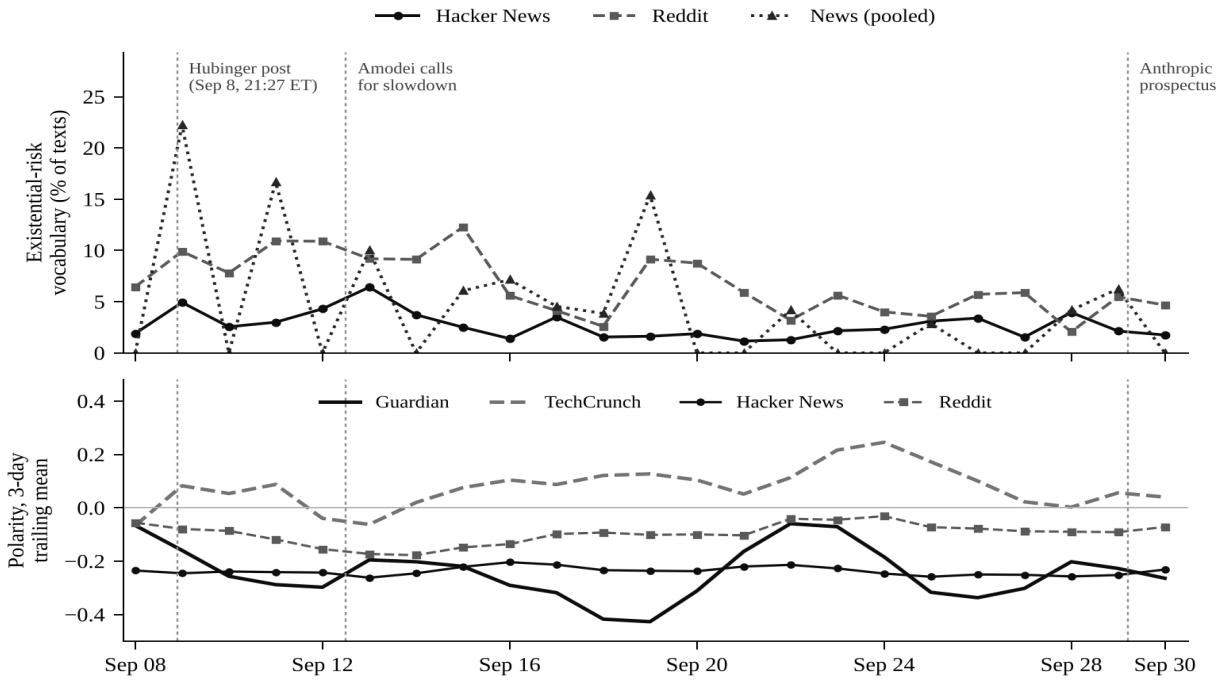


Figure 1. Top: share of texts using existential-risk vocabulary each calendar day (news pools Guardian and TechCrunch headlines). Bottom: mean RoBERTa polarity, count-weighted 3-day trailing average. Dashed lines mark the Hubinger post, the Amodei slowdown call and the Anthropic prospectus.

Table 2. Existential vocabulary (% of texts) in the five days after the post versus September 14-30.

Source	Sep 9-13	Sep 14-30	Difference, pp [95% CI]	Holm p	Polarity diff. (Holm p)
Hacker News	4.2%	2.3%	+1.9 [+0.6, +3.3]	0.015	-0.022 (0.240)
Reddit	9.7%	5.7%	+4.0 [+2.2, +5.8]	<0.001	-0.058 (0.057)
News headlines	9.8%	3.2%	+6.6 [+1.3, +11.8]	0.016	-0.071 (0.240)

Daily means with lag-one HAC intervals; Holm adjustment across the three sources within each metric.

Figure 1 shows the clearest trend in the data. On Hacker News the share of AI comments using existential vocabulary rose from 1.9% on September 8 to 4.9% on September 9 and peaked at 6.4% on September 13; threads titled "Anthropic researcher says more than 10% chance AI could kill all humans" and "I resigned from Anthropic today" dominated the discussion. The five-day post-event average was 4.2% against 2.3% for the rest of the month, a gap of +1.9 points (Table 2). Reddit, already discussing the resignation on September 8, stayed elevated until September 15 as follow-on stories (a Senate bill to ban superintelligent AI, a Google researcher quitting with a similar warning) kept the theme alive, then fell back apart from a brief rise on September 19-20. News headlines spiked as well: three of the Guardian's seven AI headlines on September 9 concerned extinction. The daily X search sample shows the same pattern: 44% of posts used risk vocabulary in September 8-13 against 22% later (n = 18 and 51). In GDELT's English web news, AI-risk articles were 15.4% of AI coverage on September 10 against 3.1% on September 19. Attention decayed within about a week and rose modestly again around the September 29 prospectus.

Overall polarity, by contrast, hardly moved. Daily linear trends are indistinguishable from zero for every source (Hacker News -0.0001 per day, p = 0.92; Guardian p = 0.97; TechCrunch p = 0.27), and the event-week polarity differences in Table 2 are small. The warning changed what people talked about more than how they felt about AI in general: only a few percent of AI comments discuss existential risk, and those that do were already strongly negative. Within that subset, the Hacker News negative share was 59.9% in September 9-13 against 54.8% later, so the risk conversation itself became somewhat more negative while it lasted.

5. The AI equity basket

The basket holds NVDA, MSFT, AMZN, GOOGL, META and AVGO with equal daily weights. It spans accelerators and networking (NVDA, AVGO), cloud and enterprise AI (MSFT, AMZN, GOOGL) and frontier models in consumer products (META); constituents were fixed before any returns were examined, following the defined-basket approach of Aloosh, Choi and Ouzan (2023). Anthropic itself is private. Returns are close-to-close on adjusted prices, with September 4 as the base because September 7 was Labor Day.

Table 3. Cumulative returns, September 8-30.

NVDA	MSFT	AMZN	GOOGL	META	AVGO	Basket	Excl. META	SPY	QQQ
-0.75%	2.64%	-3.62%	1.66%	17.67%	-1.70%	2.72%	-0.25%	-0.74%	3.00%

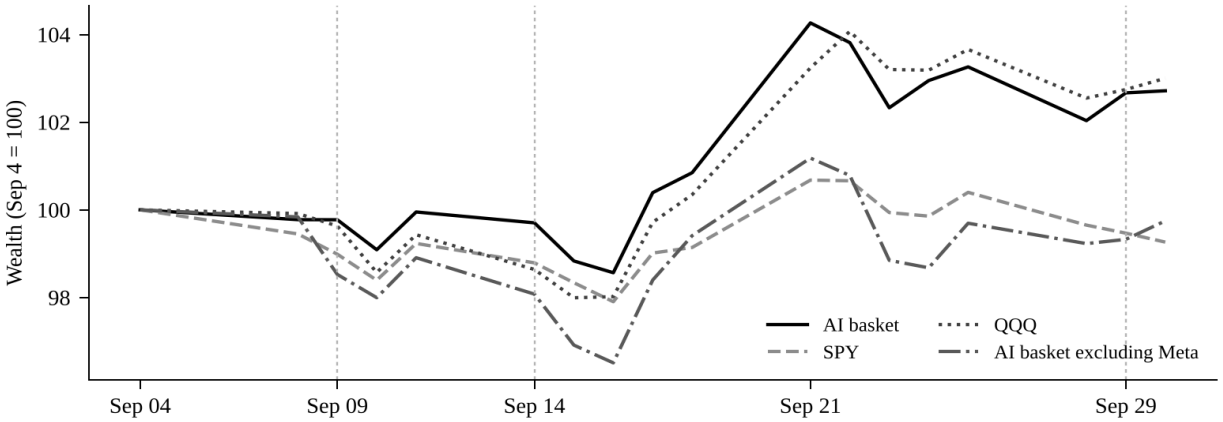


Figure 2. Compounded wealth, September 4 = 100. Dashed lines mark September 9, 14 and 29.

The basket gained 2.72%, beating SPY by 3.45 points but trailing QQQ by 0.28. Meta, whose Muse agent launched on September 8 and reached No. 2 in the US app rankings, rose 17.7%; dropping any other single stock leaves the basket return between 2.7% and 4.0%, whereas dropping Meta turns it to -0.25%. Table 4 isolates the sessions that followed the three main risk events.

Table 4. Daily returns (%) in sessions after risk events.

Session	Preceding event	Basket	Chips	Platforms	QQQ	Basket − QQQ
Sep 9	First session after the Hubinger post	0.00	-1.02	+0.51	-0.29	+0.28
Sep 14	First session after the Amodei slowdown call	-0.25	-4.06	+1.66	-0.80	+0.55
Sep 29	Session after the Anthropic prospectus	+0.62	+0.43	+0.72	+0.19	+0.43

Chips = NVDA, AVGO; platforms = MSFT, AMZN, GOOGL, META. Over the 17 sessions the daily standard deviation is 0.47 points for basket − QQQ and 1.12 for chips − QQQ.

The September 9 session shows no broad sell-off: the basket was flat (0.00%), as Meta's +6.6% gain on its agent launch offset a 1.02% fall in the chip names, and QQQ and SPY dipped slightly. The sharpest reaction came on September 14, the first session after Dario Amodei's weekend call to slow AI development, which the Guardian reported as "AI-linked stocks slide": the two semiconductor names fell 4.06% on average against -0.80% for QQQ, while the platform companies rose. A call to slow development threatens chip demand directly, whereas an abstract extinction estimate does not. The chip names trailed QQQ by 3.27 points, about 2.9 standard deviations of their daily spread; with a single event this is suggestive rather than conclusive.

6. Granger causality

Each text is assigned to the first regular close at or after its publication, so weekend and after-hours texts join the next trading window. For a sentiment series S_t on this trading grid I estimate

$$r_t = \alpha + \varphi r_{t-1} + \beta \Delta S_{t-1} + \varepsilon_t,$$

where r_t is the basket return in percent, and test H_0: $\beta = 0$ with the nested-model F test. Sentiment changes are used because the assignment asks about changes in sentiment; the lag order is fixed at one in advance, given only 17 sessions. News sentiment is the 50/50 average of the two outlets, so a busy day at one publisher cannot shift the index. The two primary tests (news and Hacker News tone) are Holm-adjusted together; the four supplementary tests in the lower rows of Table 5, added to follow the risk-attention finding of Section 4, form a second family. The first usable response is September 10, since one lagged change requires two prior windows.

Table 5. Granger tests of ΔS_{t-1} on the basket return.

Predictor ΔS	N	β×0.1 [95% CI]	F	p	HC3 p	Holm p
News tone (primary)	15	-0.44 [-1.26, +0.37]	1.41	0.259	0.259	0.517
Hacker News tone (primary)	15	-1.37 [-3.97, +1.23]	1.32	0.273	0.115	0.517
Reddit tone change	15	+0.34 [-1.15, +1.84]	0.25	0.628	0.726	1.000
HN risk-attention change	15	-1.43 [-6.92, +4.06]	0.32	0.580	0.525	1.000
Social risk-attention change (HN+Reddit)	15	-2.19 [-8.02, +3.64]	0.67	0.429	0.304	1.000
News risk-attention change	15	+0.12 [-1.11, +1.35]	0.04	0.837	0.887	1.000

Upper rows: primary family; lower rows: supplementary family. β×0.1 is the next-day return response (percentage points) to a 0.1 rise in the change of polarity or of the share of texts with existential vocabulary.

No null hypothesis is rejected. The point estimates for tone are negative (a rise in polarity is followed by slightly lower returns), the opposite of a "bad news depresses prices" story, and their intervals comfortably include zero. With twelve residual degrees of freedom the test has little power: an effect would need to be large to be detected. The result is robust to the alternatives in the accompanying code: two lags, tone levels instead of changes, returns in excess of SPY or QQQ, a lagged SPY control, alternative weighting, the Hu-Liu measure and the explicit-AI headline subset. Across these 19 secondary tests the smallest p-value is 0.051 (news tone with Meta removed from the basket), and none survives Holm adjustment; the VADER versions of the primary tests give p = 0.43 and 0.69. The reverse regression of sentiment changes on lagged

returns is also insignificant (news p = 0.33, Hacker News p = 0.51). Smailović et al. report p = 0.0565 for two lags on Apple tweets; with a far shorter sample, I cannot detect even an effect of that order.

7. Is AI-risk coverage biased?

Table 6. Tone and topic by source, September 8-30.

Source	N	Negative	Mean polarity	Risk vocab.	Existential vocab.
Guardian headlines	226	36.7%	-0.256	19.0%	8.0%
TechCrunch headlines	262	11.1%	+0.091	5.7%	1.9%
Hacker News comments	42,486	39.1%	-0.237	10.4%	2.8%
Reddit, AI/tech subreddits	3,761	22.7%	-0.112	17.6%	6.9%
Reddit, investor subreddits	570	18.1%	-0.008	18.6%	5.6%
GDELT AI-risk news, US outlets	1,279	83.5%			
GDELT AI-risk news, non-US outlets	920	62.4%			

Negative = RoBERTa negative class, or for GDELT an article in a negative bin of GDELT's own tone measure (not comparable in level with RoBERTa).

There is a measurable bias in the sense of the Covid study: sources describing the same three weeks differ systematically in both tone and topic selection. Comparing the Guardian and TechCrunch on the same calendar days (23 days, lag-one HAC, Holm-adjusted over twelve source-by-metric contrasts), Guardian headlines are 0.28 polarity points more negative (Holm p < 0.001), carry a negative label +20.3 percentage points more often [+6.6, +34.0] (Holm p = 0.045), and use risk vocabulary +11.1 points more often (Holm p = 0.026). The framing difference is visible in the headlines themselves. On September 9 the Guardian ran "AI could kill all humans in next decade, warn experts", and later "How would AI actually 'kill all humans'? Here are the top five most likely scenarios"; TechCrunch's main follow-up was "An Anthropic researcher's doomsday warning comes at a very interesting time", which questions the timing of the warning rather than its content. Hacker News comments are as negative as Guardian headlines, but in the coded sample that negativity mixes fear of AI with skepticism toward the warning itself, often dismissed as hype or a bid for regulatory capture.

The closest analogue to Sacerdote et al.'s comparison of US and non-US outlets comes from GDELT. Among English-language web articles that mention artificial intelligence together with extinction-type terms, 83.5% of US-sourced articles fall in a negative tone bin against 62.4% of non-US articles, a gap of +21.1 points [+17.4, +24.8]. Sacerdote et al. report 91% versus 54% for Covid coverage, so the direction matches. Syndicated copies of the same story mean the articles are not independent, so the interval understates the uncertainty. On Reddit, investor communities are less negative than the AI and technology communities (18.1% versus 22.7%; day-level difference -5.7 points, p = 0.007), and the market itself sent a different message: the AI basket gained 2.7% while most risk coverage was negative.

Table 7. Direct reactions to the Hubinger post on X (first 20 quotes and 20 replies, author's own reply and one verbatim copy excluded).

Stance	N	Share	Share of likes	RoBERTa negative
Alarm or call to stop	7	18%	32%	57%
Moral challenge to lab staff	8	21%	24%	88%
Skepticism (hype, capture)	8	21%	9%	62%
Humor or irony	8	21%	13%	12%
Analysis, news or other	7	18%	22%	57%

The reactions on X itself show where the negativity of social media comes from. Of 38 direct reactions to the post, 55% are classed as negative, and 77% when each is weighted by its likes. Coded by stance (Table 7), the reactions split almost evenly between alarm, moral challenges to the people who keep building ("what kind of person keeps building something they believe is a civilization destroyer?"), skeptical dismissal as hype or regulatory capture, and humor. Engagement is not even, however: alarm and moral challenges collect 57% of all likes, skeptical replies only 9%. The most visible reaction on X therefore amplified the warning, whereas on Hacker News much of the negativity was directed at the warning itself. Because X ranks replies by relevance and the sample is small, this is descriptive evidence only.

Two limits apply. Sacerdote et al. benchmark coverage against an objective series, case counts, which rose and fell while coverage stayed negative. No comparable daily measure of true AI risk exists, so I can show that coverage is more negative and more risk-focused in some outlets than in others, but not that any of them overstates the actual danger. Negativity may also reflect audience demand, which is part of Sacerdote et al.'s explanation for US outlets.

8. Conclusion

The Hubinger post produced a clear but short-lived spike in attention to existential AI risk across forums and news, followed by a second rise around Anthropic's prospectus, while general sentiment toward AI stayed flat. The AI basket rose 2.72%, almost entirely because of Meta, and neither sentiment changes nor risk-attention changes Granger-cause its returns over seventeen sessions; the most visible market response was a one-day drop in chip stocks after the call to slow AI development. Coverage of AI risk is biased in a measurable, source-dependent way, with general-interest and US outlets markedly more negative than trade and non-US outlets. The main limitations are the short sample, which leaves the Granger tests with low power, headline-level news text, and an X sample that is small and hand-collected.

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